

Shantanu Gorai

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SUMMARY

Data Science and ML practitioner with end-to-end ownership of applied projects spanning EDA, preprocessing, feature engineering, model selection, evaluation, and Flask-based API deployment. Strong fundamentals in supervised and unsupervised learning, anomaly detection, time-series modeling, and behavioral analytics using Python, Scikit-learn, Pandas, and NumPy. Experienced in diagnosing overfitting, handling class imbalance, running structured experiments, and maintaining reproducible, documented ML workflows. Industry-trained at Tata Steel; national-level hackathon finalist.

EXPERIENCE

Vocational Trainee — Image Recognition using Machine Learning

Apr 2026 – May 2026

Tata Steel Ltd., Learning & Development Department

Jamshedpur, Jharkhand

- Built an end-to-end image classification pipeline: preprocessing (normalization, augmentation), feature extraction, model training, and evaluation using Precision, Recall, F1, and Confusion Matrix.
- Applied train-validation-test splits and cross-validation to diagnose overfitting; documented dataset characteristics, architecture choices, and experimental results for reproducibility.

TECHNICAL SKILLS

ML & Modeling:	Supervised & Unsupervised Learning, Classification, Regression, Clustering, Anomaly Detection, Time-Series Analysis, Behavioral Modeling, Hyperparameter Tuning, Cross-Validation, Regularization
Libraries:	Scikit-learn, NumPy, Pandas, Matplotlib, Seaborn, Scapy
Feature Engineering:	Feature Selection, Dimensionality Reduction, Class Imbalance Handling, Encoding, Normalization
Evaluation Metrics:	Precision, Recall, F1, ROC-AUC, Confusion Matrix, MAE, MSE, RMSE, R^2
Deployment & APIs:	Flask, REST APIs (model serving & inference), Git/GitHub, SQL, PostgreSQL, MongoDB
Languages:	Python (primary), SQL, JavaScript, TypeScript
Data & Analytics:	EDA, Data Cleaning, Outlier Detection, Sampling Bias Identification, Experiment Documentation

PROJECTS

Real-Time Network Intrusion Detection System (IDS) | *Python, Scikit-learn, Flask, Scapy, REST API, WebSockets* 2025

- Performed EDA on network telemetry to identify traffic patterns, class imbalance, and anomalous feature distributions; engineered features from raw packet attributes (protocol, port, payload size, inter-arrival time).
- Trained and evaluated a supervised classifier using Precision, Recall, F1, ROC-AUC, and Confusion Matrix; established baselines and compared multiple algorithms before selecting final architecture.
- Deployed trained model as a Flask REST API serving real-time inference, streaming live threat predictions to a monitoring dashboard via WebSockets.

AI Student Productivity & Behavioral Analytics Dashboard | *Python, Scikit-learn, Pandas, PostgreSQL, Flask* 2025

- Preprocessed multi-source time-series behavioral data (timetables, mood logs, sleep records); handled missing values, outliers, encoding, and normalization for ML pipeline readiness.
- Applied clustering (unsupervised segmentation) and time-series modeling to predict student stress risk; ran structured experiments across algorithm/feature combinations with documented methodology and results.
- Exposed predictions via Flask REST API integrated with a React/Next.js frontend; used PostgreSQL for structured data persistence.

Geographic Velocity Anomaly Engine (Impossible Travel Detector) | *Python, Flask, IP Geolocation API, Risk Scoring* 2025

- Designed an anomaly detection and algorithmic risk-scoring system analyzing user authentication behavioral data — computing geographic velocity between login events to flag impossible travel sequences.
- Enriched data via live IP Geolocation APIs; applied iterative threshold calibration with documented edge-case analysis; exposed the engine as a production-style REST API with structured alert output.

EDUCATION

Bachelor of Engineering – Computer Science

2023 – 2027

ACS College of Engineering, Bengaluru (VTU Affiliated)

Bengaluru, Karnataka

ACHIEVEMENTS

- **Runner-Up, CODE CARNAGE – National Hackathon** (Apr 2025): Built *VocalLock*, an AI Voice Authentication & Spoof Detection system combining signal processing and ML classification in 24 hours.
- **Semi-Finalist, CODE BREAKER Challenge** (May 2025): Demonstrated structured ML problem-solving and implementation skills in a competitive, time-bound environment.
- **Top-5 Finalist, AI-VERSE 2.0** (Oct 2025): Designed, evaluated, and presented an end-to-end AI solution among nationally competing teams.